Saturated orbit forests and the even-degree boundary

Human Compression project: a component of the L019 recovery

Throughout, $G = \langle \rho_0, \dots, \rho_{r-1} \rangle$ is transitive on n points, its involutions satisfy the string relations, its noncommutation diagram is connected, and all maximal omissions are intransitive. A colour is eligible when its edge joins different orbits of its actual maximal omission. The acquired unique-crossing lemma identifies a colour with one eligible edge as a bridge between precisely two omission orbits. A perfect split has all earlier marks supported on one side and all later marks supported on the other.

Lemma 1 (A saturated transitivity forest)**.** *Let $s_1, \dots, s_t$ be an inclusion-minimal family of involutions transitive on each of two disjoint sets of sizes m_1, m_2. If $t = m_1 + m_2 - 2$, each s_l is a single transposition on one of the sets, and the corresponding edges form a tree on each set.*

Proof. For each member choose an edge between two orbits obtained by deleting that member. As in the acquired minimal-transitivity proof, the selected edges form a forest. The assumed edge count forces exactly two components, the two sets in the statement. For any further transposition of s_t, join its endpoints by the unique selected path. If this path contains an edge of some different colour l, the further t-edge and the rest of the path give an alternative path between that selected l-edge's endpoints, using no colour l. This contradicts its eligibility under deletion of s_l. The path can therefore contain only the selected t-edge; the further edge is that same edge. This applies to every member and every one of its transpositions. □

Theorem 2 (Saturation on two imprimitive sides)**.** *Suppose a unique i-edge joins $a \in A$ to $b \in B$, and G_i preserves respectively $m_1, m_2 \geq 2$ blocks of sizes $k_1, k_2 \geq 2$ on these two orbits. If $2r = n + 2$, then this cut is perfect.*

Proof. Choose an inclusion-minimal subfamily M of the omission marks transitive on both block sets. The full orbit-budget proof gives, with c_j its number of point orbits on side j,

$$|M| \leq m_1 + m_2 - 2, \qquad c_j \leq k_j, \qquad r - 1 - |M| \leq c_1 + c_2 - 2.$$

Together with $m_1 + m_2 + k_1 + k_2 - 3 \leq (n+2)/2 = r$, these force equality in every displayed inequality. Thus $|M| = m_1 + m_2 - 2$, $c_j = k_j$, and there are $k_1 + k_2 - 2$ remaining omission marks.

Lemma 1 applies to the block images of M. The two block trees are paths: three incident transpositions would pairwise fail to commute, whereas three distinct integer labels cannot be pairwise consecutive. Consecutive edges of each path have consecutive labels; the direction cannot change without repeating a label. The block containing a is an endpoint of its path. Otherwise its two incident labels would both belong to $\{i-1, i+1\}$, because all more distant marks fix a. These two labels are distant and must commute, a contradiction. Its unique incident label is $i-1$ or $i+1$. The same reasoning applies at b, and these two incident labels are different because each member of M acts on just one block set. After reversing the order if necessary, the two block paths therefore have labels

$$i - m_1 + 1, \dots, i - 1 \quad \text{on } A, \qquad i + 1, \dots, i + m_2 - 1 \quad \text{on } B,$$

with the crossing block at the endpoint next to i. These are all the members of M.

Each point orbit of $\langle M\rangle$ meets every block of its side. Since there are exactly k_j such orbits, each has exactly m_j points and meets each block once. Projection from any such orbit to the block set is an equivariant bijection. In particular the block action of $\langle M\rangle$ is faithful; its marks act as the natural path transpositions on each of these copies. A mark in the first block path fixes all points of B, and conversely for the second path.

Let t be a remaining omission mark. Its label is outside the entire interval occupied by $M \cup \{i\}$, so it fixes a, b. On either block set it centralizes the full natural symmetric action of M, except possibly for the transposition at the far endpoint of that path, if its label is adjacent to t. We claim its block image is the identity in every case. For the full path, a permutation centralizing a transitive action and fixing one point is trivial. For the path with its far-end edge deleted, the centralized group is the natural S_{m_j-1} on the initial $m_j - 1$ blocks. When $m_j \geq 3$, its unique nontrivial orbit is preserved by every centralizer. Fixing the crossing block and centralizing that transitive orbit forces fixation of the whole orbit; the one remaining block is fixed too. When $m_j = 2$, fixing one of the two blocks already forces the same conclusion. This proves the claim, without asserting any action on the point-orbit set.

Now select one eligible actual edge for every remaining omission mark and contract the point orbits of $\langle M\rangle$. The acquired orbit-forest argument gives exactly two trees, one on the k_1 orbits in A and one on the k_2 orbits in B. Because t fixes every original block and each $\langle M\rangle$-orbit meets that block just once, every nontrivial transposition of t joins two distinct contracted vertices. Let u, v be such a pair. If the selected path from u to v contained the selected edge of a colour $l \neq t$, that path, the actual t-edge, and paths within $\langle M\rangle$-orbits would connect the endpoints of the selected l-edge in G_l. This contradicts actual ambient eligibility. Thus u, v must be the endpoints of the one selected t-edge. Every transposition of t consequently lies on the same side, A or B.

We have proved that every omission mark acts on precisely one side. This side cannot change at consecutive positions on either side of i, since the corresponding marks would then have disjoint supports and commute, disconnecting the noncommutation diagram. The neighbours $i-1, i+1$ are already in their respective block paths. Every earlier mark therefore acts only on A, and every later mark only on B. The split is perfect. □

Lemma 3 (The two support interfaces at the boundary)**.** *Assume $2r > n+1$ and a unique cut is not perfect. Its two sides are nontrivial, one primitive side has its active labels in an interval next to the cut, and after reversal $G_{<i}$ fixes that side. The acquired head-support theorem, in this orientation, remains valid: for the last shared index $h > i$,*

$$h < r - 1, \qquad X = \operatorname{supp}(G_{>h}) \subseteq B \setminus \{b\}, \qquad 2h \leq n - |X| - 1,$$

with the endpoint interchange allowed there.

Proof. The acquired orbit budget gives $2r \leq n + 2$ when both sides are imprimitive. Under the present strict inequality, its only possible equality boundary is $2r = n + 2$; that cut is perfect by Theorem 2. Thus a nonperfect cut cannot have two imprimitive sides. An interior singleton would give $2r \leq n + 1$, while an endpoint singleton is already perfect. The remaining primitive-side argument is exactly the acquired argument: disconnected active indices force two commuting transitive factors, hence the exceptional degree-two action of the two neighbouring involutions. Its acquired bound is $2r \leq n$. It is excluded here too. This gives the stated interval and orientation.

The first five lemmas of the head-support manuscript have no rank-range assumption. The fifth gives $n \geq 2r$ when a final shared mark, or a final mark on the wrong side of a positive cut, occurs. Both are excluded by $2r > n+1$. Connectedness then puts the final tail on a single side. The rest of that theorem's proof is a count independent of the rank range: two fixed path pieces at $i = 0$; at $i > 0$ either $|A|+1 \geq 2i+3$ for a transitive left side, or $|A|+(h-i) \geq 2h+1$ from its commuting-orbit and internal-pair lemmas. All these inequalities were proved with their weak

equality cases retained. They give the claimed head bound and its support placement under the present hypothesis. □

Theorem 4. *Under the standing hypotheses,* $2r > n + 1$ *implies the existence of a perfect split.*

Proof. Use the complete acquired two-support proof with Lemma 3 supplying its orientation and head bounds. We specify every place affected by the changed rank assumption. The all-paired case gives $2r \leq n$. The first tail having all colours paired gives $2r \leq n + 1$. The second-cut orientation uses only the last-mark bound $n \geq 2r$, and the separated-region count excludes $g \leq h$ by $n \geq 2r + h - g$. All are contradictions to the present hypothesis. Thus that proof reaches exactly the same actual sets X, Y and indices $h < g$, with

$$X \cup Y = \Omega, \quad 2h \leq n - |X| - 1, \quad 2(r - 1 - g) \leq n - |Y| - 1.$$

If $g \geq h + 2$, its middle interval is nonempty and intransitive. Retain the full ambient interval inequality, including its positive orbit deficit $c \geq 1$:

$$2(g - h - 1) \leq |X \cap Y| - c.$$

Summing the three bounds now gives $2r \leq n + 2 - c \leq n + 1$.

If $g = h + 1$, there is no middle interval. The mark ρ_h is in the group defining Y, and ρ_{h+1} is in the group defining X. They fail to commute by connectedness of the string noncommutation diagram. Their supports must intersect; in particular $|X \cap Y| \geq 1$. Sum just the two outside bounds, retaining this overlap:

$$2r = 2h + 2(r - 1 - g) + 4 \leq 2n - |X| - |Y| + 2 = n - |X \cap Y| + 2 \leq n + 1.$$

This is the same contradiction and proves the theorem. □

Attribution and scope. The block systems and the two-support route originate in Cameron–Fernandes–Leemans and their earlier alternating-group work. The full actual orbit forests, paths and ambient interval proofs are acquired premises. The saturated transitivity argument above replaces a classification of equality diagrams: it controls full actions on the original blocks before using the contracted orbit forest, and never assumes that a quotient marking has IP. The positive middle deficit and the adjacent support overlap pay the one-unit boundary separately.

For an actual wing of a parent marking whose maximal omissions are intransitive, every maximal omission inside that wing is intransitive too. Indeed a transitive wing omission, together with the other full wing and the crossing mark, would make the corresponding parent omission transitive. Thus the theorem is available for an even-degree symmetric wing of rank $m/2 + 1$, provided its other stated hypotheses hold. It does not by itself supply an internal split with a trivial internal factor, the leaf-kernel conditions, or the stability bijection. Those obligations and the complete original remain uncredited.